

Physics exam-15 — Marking Report

Units 1-6 and 9 · 46 marks, no optional questions · attempted 10 September 2026 · independently deep-checked

45 out of 46 · 97.8%	29/30 MCQs · 97%	16/16 Sections B + C
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The short version. Seemab scored 45 out of 46. This is her strongest paper of the year, and the reason matters more than the number.

Three weeks ago her multiple choice was the clear weak point of her work. On Maths exam-17 it was 60%. On Chemistry exam-21 it was 83%. On this paper it is **97%**. She lost exactly one mark on Section A, and even that one she knew the answer to.

The change is visible on the page. This is the first attempt where she has written out working and a stated reason for her multiple choice answers instead of just letters. Question 1 explains why the leading zeros in 0.02080 do not count. Question 6 explains why the body is at rest rather than accelerating. Question 16 explains why the odd one out is a disadvantage rather than an advantage. That was the single piece of advice given to her, and she acted on it.

All four repeat errors: corrected. Four questions on this paper were chosen deliberately because they are the exact multiple-choice questions she has got wrong on earlier papers. She answered all four correctly, three of them with working shown.

Q	Her old mistake	This time
7	Gradient of a speed-time graph, wrong on exam-12 and again on exam-14	$a = (40-10)/15 = \mathbf{2\ m/s^2}$ ✓
8	Area under a speed-time graph, her longest-standing physics error	distance = $12 \times 25 = \mathbf{300\ m}$ ✓
19	The SI unit of pressure, wrong on exam-12	$\mathbf{1\ N/m^2}$ ✓
24	Renewable vs non-renewable, wrong on exam-14	Geothermal ✓

Gradient against area was the pair she had never separated. She got both directions right, in the same paper, with the arithmetic written out.

The one mark lost, and it is a frustrating one

Question 3 asked which instrument measures weight directly. The options were A physical balance, B spring balance, C measuring cylinder, D vernier caliper. **She wrote A.**

But her own written reason underneath says, word for word: *"because spring balance is measuring force, measuring cylinder is used to measure volume and vernier calipers are used to measure small lengths."*

She identified the spring balance as the force-measuring instrument and correctly ruled out two of the other three. She reasoned her way to B and then recorded A. **The mark is lost to transcription, not to knowledge.**

This is worth saying to her directly, because it is now the second paper running where a mark went this way: on Maths exam-17 she wrote a coordinate instead of an option letter and it could not be read. **The habit to build is simple: after writing the reason, look back at the options and check the letter you have written actually matches the one you argued for.**

Section A — 29/30

Twenty-eight letters correct, one wrong (Q3), and one question answered by working alone (see below). A shift guard was run to rule out a mis-numbered answer list: 29 matches at the correct alignment against 1, 1, 3 and 1 at the four neighbouring offsets. No misalignment.

Q	Topic	Her working, quoted
1	Significant figures	"ignoring first two zeros we have 2080"
5	Free fall	$t = (35 - 0)/10 = 3.5\ \text{s}$
9	Impulse	$I = (0.5)(0) - (0.5)(20) = -10\ \text{N}\cdot\text{s}$
10	Conservation of momentum	$0.01 \times 200 + 1.99 \times 0 = 2$, so $v = 1\ \text{m/s}$
13	Principle of moments	$30 \times 2 = F_2 \times 3$, $F_2 = 20\ \text{N}$
17	Manometer	$P = \rho gh = 1200 \times 10 \times 0.15 = 1800\ \text{Pa}$
20	Hydraulic press	$F_2 = 20 \times 0.45 / 0.0015 = 6000\ \text{N}$
22	KE and PE conservation	$KE = PE = mgh = 4 \times 10 \times 10 = 400\ \text{J}$
23	Work at an angle	$W = 90 \times 3 \times \cos 60^\circ = 135\ \text{J}$

Question 18 needs a decision. She wrote **no option letter** at all, only the working: $F = kx$, $k = 40 / 0.02 = \mathbf{2000\ N/m}$. That value is correct and matches exactly one option, C.

I have **awarded** the mark, on the rule that a written letter is the authoritative answer when one exists, and the working governs only

when no letter was written. Under that same rule Q3 stays wrong, because there she did write a letter.

Note the precedent: on Maths exam-17 Q10 she also wrote a value instead of a letter and it was ruled wrong. The difference relied on here is that Q10 was genuinely illegible, while this is unambiguous and correct. **If you would rather be consistent with that ruling, the paper is 44/46 (95.7%).**

Section B — 8/8

Q	Topic	Mark	Comment
2(i)	Theory vs law	4/4	Full comparison table. Theory explains <i>why</i> and can be modified with new evidence; law describes <i>what</i> happens, often mathematically. Examples: Cell Theory and Atomic Theory against Hooke's Law and Newton's Law of Universal Gravitation. Her examples differ from the key's suggestions and are equally valid.
2(ii)	Weight on Earth and Moon	4/4	(a) $55 \times 9.8 = 539 \text{ N}$. (b) $55 \times 1.62 = 89.1 \text{ N}$. (c) <i>"the mass does not change because mass is the amount of matter in an object and remains constant regardless of the gravitational field. Only the weight changes because the moon has a weaker gravitational field than earth."</i> Complete, with the reason stated.

Both were items her school examined that had never appeared on any of our fourteen previous physics papers.

Section C — 8/8

The full satellite orbital numerical, which the coverage tracker had flagged as the longest-standing gap in Unit 4.

Part	Required	Her answer	Mark
(a)	Define orbital motion, state the speed formula	Correct, $v = 2\pi r/T$	2/2
(b)	Orbital speed in m/s	7851 m/s	2/2
(c)	Convert to km/h	28,264 km/h	1/1
(d)	Name the force and its direction	Gravitational, towards the centre of the Earth	1/1
(e)	Centripetal force on a 750 kg satellite	$6.80 \times 10^3 \text{ N}$	2/2

On (b) the exact value is 7854 m/s. Her 7851 is a small unshown-work slip in the long division, about 0.05% out; no standard value of π produces 7851 exactly. The method and setup are perfect and the deviation is immaterial, so full marks stand. Her (c) and (e) were then checked against *her own* figure rather than the ideal one: $7851 \times 3.6 = 28,264$, and $750 \times 7851^2 / 6,800,000 = 6798 \text{ N}$, which rounds to the $6.80 \times 10^3 \text{ N}$ she wrote. Both are genuinely consistent.

What to say to her

- **The rough working worked.** 60% to 83% to 97% in four days, on the one thing she was asked to change. She should hear that plainly.
- **Gradient and area are fixed.** The error she had repeated across two papers did not appear, in either direction.
- **Check the letter against the reason.** The single lost mark was a question she had already reasoned correctly. Ten seconds of checking would have made this a perfect paper.
- **Always write the option letter,** even when the working makes the answer obvious. On Q18 it happened to be recoverable. It has not always been.

Marking independently deep-checked: all 30 MCQ answers re-derived from the printed question stems rather than the answer key (no key error found), her answers re-read letter by letter, a shift guard run across five alignments, and every calculation in Sections B and C re-worked by hand. Both passes reached 45/46 with no disagreement on any question.